

Data Mining (CT72502 / CT725)

Detailed PYQ

August 11, 2026

Notes

- Questions from Tribhuvan University, IOE (2070–2081 BS), sorted by the CT72502 syllabus.
- Regular exam = **bold**, Back / New Back exam = regular font.
- Data Mining is a *single common paper* for both BEX and BCT (Elective I / Elective II), so there is no separate programme axis — no **monospace** style is used in this document.
- So: **80 Bh** = Regular, 81 Ba = Back.
- Months: Ba = Baishakh, Shr = Shrawan, Bh = Bhadra, Ash = Ashwin, Ka = Kartik, Ch = Chaitra.
- Keywords: (I) = Importance/Need/Significance, (+) = Advantage, (-) = Disadvantage, +Fig = Figure/Diagram, +Eg = Example, Vs = Compare/Differentiate.
- Papers covered (17): **80 Bh**, 81 Ba, 80 Ba, **79 Bh**, **78 Bh**, **76 Ch**, 76 Ash, **75 Ch**, 75 Ash, **74 Ch**, 74 Ash, 73 Shr, **72 Ch**, 72 Ka, **71 Ch**, 71 Shr, **70 Ch**. There was no exam in 2077; 2073 Ch, 2070 Ash and 2069 are absent from the source compilation.
- **71 Ch** and 71 Shr print no per-question marks (“All questions carry equal marks”, 80 marks total).

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1 Introduction

(2 Hours)

1.1 Data Mining Origin

1. What is data mining? Explain the process of data mining. [2+3] (74 Ash)
|→ What is data mining? Explain all the steps of knowledge discovery. [2+6] (**72 Ch**)
|→ What is Data Mining? What are the steps involved in knowledge discovery process? [1+5] (**79 Bh**)
|→ What is Data mining? Explain the steps of KDD process briefly. [2+5] (81 Ba)
|→ What is a data mining? Explain general steps in brief. [4] (72 Ka)
|→ What is a Data Mining? Explain its application. [10] (**71 Ch**)
2. “The world is data rich but information is poor”. Justify with your own words. [8] (73 Shr)

1.2 Data Mining & Data Warehousing basics

1. What is data warehousing? Explain with example where data warehouses are used. [2+4] (80 Ba)
|→ Write key features of data warehouse. Explain each steps of knowledge discovery data mining process with a suitable example. [2+5] (**80 Bh**)
|→ Explain Data Warehouse architecture with its analytical processing. [8] (**75 Ch**)
2. What are the fundamental differences between Data Mining and Data Warehousing? Describe the steps of KDD for data mining. [3+7] (76 Ash)
|→ How is data warehouse different from a database? How are they similar? [2+2] (75 Ash)
3. Explain how data mining system can be integrated with database/data warehouse system. Explain Data mining process with diagram. [4+2] (**78 Bh**)
4. What is data warehouse and data mart? Describe Snowflake scheme with example. [2+4] (**74 Ch**)
5. Write short notes on: Data ware house and Data mart. [15] (**70 Ch**) [-] (71 Shr)

2 Data Preprocessing

(6 Hours)

2.1 Data Types and Attributes

1. What is data mining? Explain different data types of attributes in a dataset. [10] (71 Shr)

2.2 Data Pre-processing

1. What is Data Pre-Processing? Briefly explain the major tasks performed in data pre-processing. [2+7] (81 Ba)
|→ Why data preprocessing is necessary? Explain the methods for data preprocessing to maintain data quality. [4+4] (**75 Ch**)
|→ Why data preprocessing is required in the data mining? Explain some of approaches of data clearing. [5+5] (72 Ka)
2. What are the measuring elements of data Quality? Explain different data transformation by normalization methods with an example. [2+6] (73 Shr)
3. Use the following methods to normalize the data: 200, 300, 400, 600 and 1000. [2+2+2] (**78 Bh**)
 - a) Min-max normalization by setting min=0 and max=1
 - b) Z-score normalization
 - c) Normalization by decimal scaling
4. In real-world data, tuples with missing values for same attributes are a common occurrence. Describe various methods for handling this problem. [5] (74 Ash)
|→ What are the approaches to handle missing data? (asked together with OLAP — full statement in 2.3) [2] (**74 Ch**)
5. Discuss issues to consider during Data Integration. (asked together with OLAP — full statement in 2.3) [5] (75 Ash)
6. What is “curve of Dimensionality”? How can it be avoided? [5] (**70 Ch**)
7. Discuss the impact of noisy data in data mining? [5] (**70 Ch**)
8. How can principle component analysis be used for dimensionality reduction? [10] (71 Shr)
9. Find the principal components and the proportion of the total variance explained by each when the covariance matrix of the three random variables X_1 , X_2 , and X_3 is: [4] (**76 Ch**)

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

10. Write short notes on: Data smoothing techniques. [3] (76 Ash)
11. Write short notes on: Data Normalization. [4] (**75 Ch**)
12. Write short notes on: Data transformation. [4] (73 Shr) [3] (72 Ka)

2.3 OLAP & Multidimensional Data Analysis

1. Explain typical OLAP operations over a multidimensional data warehouse? Differentiate between OLAP and OLTP tools. [6+4] (**79 Bh**)
|→ How do you perform analysis of multidimensional data? Explain with the concept of OLAP. [10] (**72 Ch**)
|→ Discuss issues to consider during Data Integration. Describe OLAP and operations on OLAP with suitable example. [5+5] (75 Ash)

- What are the approaches to handle missing data? Describe OLAP and operations on OLAP with suitable example. Differentiate between OLAP and OLTP. [2+5+3] (74 Ch)
- What do you mean by dimensional data? What are base & apex cuboid? Slicing & Dicing? Roll Down and Roll UP operations? Give example. [2+3+3+3] (76 Ash)
 - Suppose that a data warehouse consists of the four dimensions data, spectator, location, and game, and the two measures count and charge, where charge is the fare that a spectator pays when watching a game on a given date. Spectators may be students, adults or seniors, with each category having its own charge rate. [3+3] (78 Bh)
 - Draw a star schema diagram for the data warehouse.
 - Starting with the base cuboid [data, spectator, location, game], what specific OLAP operations should you perform in order to list the total charge paid by student spectators at Dashrath Stadium in 2021?
 - Suppose that a data warehouse for a sales company consists of five dimensions: *time*, *location*, *supplier*, *brand*, and *product*, and two measures: *count* and *price*. [3+3] (76 Ch)
 - Draw a *snowflake schema* diagram for the data warehouse.
 - Starting with the base cuboid [*time*, *location*, *supplier*, *brand*, *product*], what specific OLAP operations should one perform in order to list the total *count* for a certain *brand* for each *state* per *year* (assume *location* has three levels: *country*, *state*, *city*; and assume *time* has three levels: *year*, *month*, *day*)?
 - Write short notes on: OLAP Operations. [4] (74 Ash) [4] (73 Shr)
 - Write short notes on: OLAP and OLTP. [4] (75 Ch)

2.4 Various Similarity Measures

- How do similarity / dissimilarity is calculated? Find the cosine similarity between Object-2 and 4. Also calculate the Euclidean distance between object 1, 3 and object 1, 4. [3+3+3] (80 Bh)

Object	Size	Weight	Color_Code	Taste_Score
1	4	56	7	10
2	3	53	8	11
3	7	58	6	9
4	9	55	7	12

- List down the different types of similarity measures by highlighting their application areas. [3] (80 Ba)
- Consider the following table: [2+2+2+1] (80 Ba)

Name	Gender	Eyecolor	Haircolor	Test-1	Test-2	Fever	Cough
Ram	M	Black	Gray	P	N	P	N
Laxmi	F	Blue	Black	P	P	N	N
Shyam	M	Blue	Gray	N	P	N	P

- Calculated Jaccard Coefficient of $Sim_{jaccard}$ (Ram, Laxmi) for asymmetric binary attributes.
 - Dissimilarity of symmetric binary attributes d (Laxmi, Shyam).
 - Find the Simple matching coefficients of SMC (Ram, Shyam).
 - Find the Cosine similarity between documents $d1 = (4, 1, 2, 0, 2, 0, 0)$ and $d2 = (2, 1, 3, 0, 1, 1, 1)$
- (a) Given the following points compute the distance matrix using the Manhattan and the Supremum distance. [2+1+2] (76 Ch)

Points	X	Y
P1	6	3
P2	2	2
P3	3	4

- Given the following two vectors compute the Cosine similarity between them.

$$D1 = [4 \ 0 \ 2 \ 0 \ 1] \quad D2 = [2 \ 0 \ 0 \ 2 \ 2]$$

(c) Given the following two binary vectors compute the Jaccard similarity and Simple Matching Coefficient.

$$P = [0 \ 0 \ 1 \ 1 \ 0 \ 1] \quad Q = [1 \ 1 \ 1 \ 1 \ 0 \ 1]$$

5. Explain the properties that a Distance Metric needs to support with respect to Minkowski's distance.
[10] (**71 Ch**)
6. Write short notes on: Minkowski Distance. [4] (**78 Bh**)
7. Write short notes on: Various similarity measures between data tuples. [3] (76 Ash)

3 Classification

(12 Hours)

3.1 Basics and Algorithms

1. Draw clear block diagram depicting different stages in classification. Explain the inverse relation between precision and recall. Given the confusion matrix, determine accuracy, sensitivity and precision of the classifier model. [2+3+5] (**74 Ch**)

Actual \ Predicted	Positive	Negative
Positive	142	40
Negative	98	720

2. Why is classification a supervised learning method? Explain different impurity measures used in decision tree classifier. [10] (71 Shr)
3. When do we use classifier? (asked with the KNN numerical — full statement in 3.4) [3] (**80 Bh**)

3.2 Decision Tree Classifier

1. Draw decision tree for the given data using ID3 algorithm. [10] (**79 Bh**)

Age	Income	Student	Credit_Rating	Buy's_Computer
Youth	High	No	Fair	No
Youth	High	No	Excellent	No
Middle_Aged	High	No	Fair	Yes
Senior	Medium	No	Fair	Yes
Senior	Low	Yes	Fair	Yes
Senior	Low	Yes	Excellent	No
Middle_Aged	Low	Yes	Excellent	Yes
Youth	Medium	No	Fair	No
Youth	Low	Yes	Fair	Yes
Senior	Medium	Yes	Fair	Yes
Youth	Medium	Yes	Excellent	Yes
Middle_Aged	Medium	No	Excellent	Yes
Middle_Aged	High	Yes	Fair	Yes
Senior	Medium	No	Excellent	No

2. Construct a decision tree for the following data set using information gain. Predict the class label for a data point with values <Female, 2, standard, high>. [8] (**78 Bh**)

Gender	Car ownership	Travel cost	Income level	Transport mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Female	1	Cheap	Medium	Train
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

3. The following dataset will be used to train a decision tree for predicting whether a mushroom is edible or not based on its shape, color and odor. [2+5] (**76 Ch**)

Shape	Color	Odor	Edible
C	B	1	Yes
D	B	1	Yes
D	W	1	Yes
D	W	2	Yes
C	B	2	Yes
D	B	2	No
D	G	2	No
C	U	2	No
C	B	3	No
D	W	3	No

- (a) Which attribute would the ID-3 algorithm choose to use for the root of the decision tree?
- (b) Draw the full decision tree that would be learned for the given data.
4. What is a decision tree and how information gain is used for attribute selection? Explain with example. [8] (73 Shr)
5. How do you measure the accuracy of classifiers? How do you select best root attribute in decision tree? Explain. [4+6] (76 Ash)
6. Define Decision Tree Classifier with Gini-Index with suitable example. How can you handle overfitting in Decision Tree? [6+4] (75 Ch)
 |→What is a decision tree? Explain Gini Index with suitable example. [10] (71 Ch)
7. Write about Hunt's Algorithm for Decision Tree induction. Explain the test conditions that can be used for different attribute types. [10] (72 Ka)
8. Explain decision tree with the concept of Naive base classification with appropriate example. [10] (74 Ch)

3.3 Rule Based Classifier

1. How does Rule Based Classifier work? Explain with suitable example. [7] (80 Bh)
 |→What is classification? Explain Rule-Based classification with its classification principles with suitable example. [2+8] (74 Ash)
2. Explain rule based classifier? How can CN2 Algorithm be used for rule based classification? Define "Accuracy" and "Laplace" measures used for rule evaluation. [9] (70 Ch)
3. Why is a conflict resolution strategy often necessary for rule-based classifiers? Describe the common conflict resolution strategies for rule-based classifiers. [2+4] (76 Ch)

3.4 Nearest Neighbor Classifier

1. When do we use classifier? You have the following information about the flower. Your job is to classify the given flower with SepalLength = 7, SepalWidth = 3.2, PetalLength = 4.7, and PetalWidth = 1.4. Use KNN algorithm for K = 3 with Euclidean distance matrix. [3+6] (80 Bh)

Id	Sepal_Lenth	Spal_Width	Petal_Lenth	Petal_Width	Labal
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	6	2.2	4	1	versicolor
5	6.1	2.9	4.7	1.4	versicolor
6	5.6	2.9	3.6	1.3	versicolor
7	6.7	3.1	4.4	1.4	versicolor

2. What is Nearest Neighbor Classifier? What are the main issues with this classifier? Propose another classifier that solves the issues. [1+3+4] (80 Ba)

3.5 Bayesian Classifier

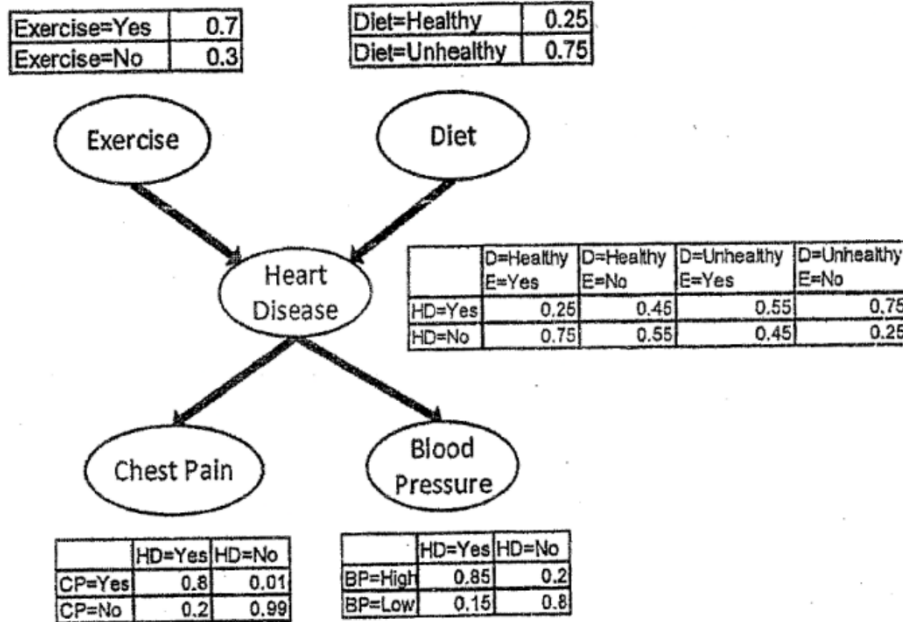
1. Explain Naïve Bayesian classification with suitable example. [8] (75 Ash)
|→ Explain Naive Bayesian classification algorithm with suitable example. [6] (**78 Bh**)
2. Suppose you have a test record “X = (Home Owner = No, Marital Status = Married, Income = \$120K)”. Your job is to classify this record using Naive Bayesian Classification. Use the following table for your calculations. [6] (**79 Bh**)

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

3. Predict Class label using naive Bayesian classifier for X = (age = youth, income = medium, student = yes, credit-rating = fair) using the following data set. [10] (**72 Ch**)

RID	Age	Income	Student	Credit-rating	Class Buy computer
1	Youth	High	No	Fair	No
2	Youth	High	No	Excellent	No
3	Middle-age	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle-age	Low	Yes	Excellent	Yes
8	Youth	Medium	No	Fair	No
9	Youth	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Youth	Medium	Yes	Excellent	Yes
12	Middle-age	Medium	No	Excellent	Yes
13	Middle-age	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No

4. What are prior and posterior probabilities? Explain the algorithmic steps of Bayesian classifier and write its strengths. [3+7] (76 Ash)
5. Explain a Bayes classifier. In what cases can Naive Bayes and Bayesian Belief Network be used? [10] (**71 Ch**)
6. Explain Naive Bayes classifier. How can over fitting problem be solved in case of classification? [10] (71 Shr)
7. What is limitation of Naive Bayes and how Bayesian Belief Networks overcomes it? If a person does exercise, eats an unhealthy diet and has blood pressure but no chest pain, will that person has a heart disease? [3+5] (81 Ba)



8. Write short notes on: Laplacian Correction in Classification method.

[4] (78 Bh)

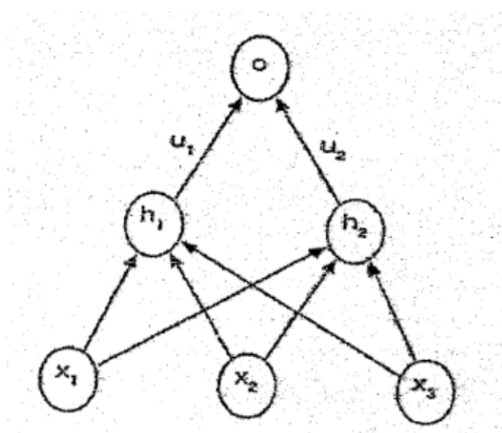
3.6 Artificial Neural Network Classifier

- How does Neural Net work classified work? Explain with suitable example. [8] (81 Ba)
 |→ How ANN works? Explain with Algorithm. [8] (75 Ch)
 |→ What is an ANN classifier? Explain its general consideration that required for the classifier. [2+6] (72 Ka)
- Consider the multi-layer feed-forward neural network shown in the following figure. This neural network has three inputs (x_1), (x_2) and (x_3) connected to a hidden layer consisting of two nodes (h_1) and (h_2). The weight of the edge connecting (x_i) to (h_j) is (w_{ji}). The two hidden nodes are connected to the output node (o). The weight of the edge connecting the hidden node (h_i) to the output node (o) is (u_i). The activation functions at hidden and output layers is set to sigmoid function defined as follows: [2+3+4] (76 Ch)

$$\sigma(\theta) = \frac{1}{1 + \exp(-\theta)}$$

Using the target output (t), the squared error is used as the loss function at the output node, and is defined as:

$$E(o, t) = \frac{1}{2}(o - t)^2$$



- (a) Using the symbols given above, compute the activation at (h_1).
 (b) Compute the gradient of the loss with respect to the output (o).
 (c) Compute the gradient of the loss with respect to the weight (w_{12}).
 3. Write short notes on: Neural Network Classifier. [5] (80 Ba)

3.7 Issues: Overfitting, Validation, Model Comparison

1. In what cases you cannot use Accuracy for performance measure, give some examples. Assume that you have the following confusion matrix. Calculate the Classification error, Sensitivity, False alarm rate Specificity. [3+5] (80 Ba)

		Actual Values	
		True	False
Predicated Values	True	1050	250
	False	150	950

2. Calculate: Accuracy, TPR, FPR and Precision for the given confusion matrix for a classifier. [4] (78 Bh)

		Actual Class	
		Class 1	Class 2
Predicted Class	Class 1	142	40
	Class 2	98	720

3. The confusion matrix for a classifier is given as follows. Calculate: Accuracy, Sensitivity, Specificity and Precision. [10] (75 Ash)

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	21	6
	Class 2	7	41

4. The confusion matrix for a classifier is given as follows. Calculate: a) Accuracy b) Sensitivity c) Specificity d) Precision [10] (74 Ash)

		Predicted Class	
		Class 1	Class 2
Actual Class	Class 1	25	9
	Class 2	4	31

5. The confusion matrix for a classifier is given as follows. Calculate a. accuracy b. sensitivity c. specificity d. precision e. recall [10] (72 Ch)

		actual class	
		class1	class2
predicted class	class1	21	6
	class2	7	41

6. Explain ROC. Using the following data, calculate TPR, FPR, precision for given confusion matrix. Classify, A = Yes, B = No. [1+3+6] (73 Shr)

	A	B
A	20	5
B	10	40

7. An input sequence “A A B B B A A A B B” was used for classification. The Classifier ‘X’ predicted the sequences as: “A A B B B A A A B B” where as the Classifier ‘Y’ predicted the sequences as: “A A A A B B A A A B”. Develop the corresponding confusion matrix for the classifiers and find their corresponding. [10] (70 Ch)

- Accuracy
- Precision
- True Positive Rate
- False Positive Rate

8. Write short notes on: Overfitting problem in classification.

[4] (78 Bh)

4 Association Analysis

(10 Hours)

4.1 Basics and Algorithms

1. What is an association analysis? Explain its importance in market-basket analysis. [2+5] (72 Ka)
|→ What is the use of Apriori Algorithm in market basket analysis? Explain with suitable example. [10] (72 Ch)
|→ Where is association analysis applicable and beneficial for us? Elaborate FP Growth Method Algorithm with examples. [2+6] (80 Ba)
2. Why association analysis is required in data mining? Explain Apriori principle with example. [2+6] (75 Ash) [2+6] (74 Ch)
3. What is the importance of SUPPORT and COFIDENCE during association analysis? Explain FP-Growth method with example. [10] (72 Ch)
4. Why is pattern evaluation important in association rule mining? Explain with example the statistical based measures used for measuring interestingness of association rules. [8] (70 Ch)
5. Generally, we will be more interested in associated rules with high confidence. However, often we will not be interested in association rules that have a confidence of 100%. Why? Then specifically explain why association rules with 99% confidence may be interesting (i.e., what might they indicate)? [4] (80 Ba) — asked with the Apriori numerical, full statement in 4.2
6. Write short notes on: Market Basket Analysis. [4] (75 Ch)

4.2 Frequent Itemset Pattern & Apriori Principle

1. What is Association Analysis? Explain with different use cases? Use the Apriori algorithm to find the frequent itemsets. Assume minimum support count is 4 and confidence is 80%. [2+7] (80 Bh)

TID	Items
T100	F, A, C, D, G, I, M, P, N
T101	A, B, C, D, F, L, M, O, P
T102	B, F, H, V, J, O, P
T103	B, C, K, S, A, V
T104	L, A, F, C, E, P, M, N, V
T105	I, B, A, P, S, M
T106	F, A, C, I, B, A, P

2. Generally, we will be more interested in associated rules with high confidence. However, often we will not be interested in association rules that have a confidence of 100%. Why? Then specifically explain why association rules with 99% confidence may be interesting (i.e., what might they indicate)? Identify the candidate and large item sets of the following transaction table using Apriori algorithm with minimum support 2. [4+5] (80 Ba)

TID	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

3. Derive association rule for the following market basket transactions. Minimum support = 50%, Minimum confidence = 80%. [8] (79 Bh)

Transaction ID	Item Set
1	A,B
2	A,D
3	A,C
4	B,E
5	B,D,E
6	A,E,C

4. Consider the transaction data shown in the following table from a fast food restaurant. [5+3] (76 Ch)

Meal Item	List of Item IDs
Order:1	M1, M2, M5
Order:2	M2, M4
Order:3	M2, M3
Order:4	M1, M2, M4
Order:5	M1, M3
Order:6	M2, M3
Order:7	M1, M3
Order:8	M1, M2, M3, M5
Order:9	M1, M2, M3

There are 9 distinct transactions (Order: 1 – Order: 9) and each transaction involves between 2 and 4 meal items. There are a total of 5 meal items that are involved in the transactions. For simplicity, the meal items have been assigned short names (M1-M5). Assume that the minimum support is 2/9 and the minimum confidence is 7/9.

- (a) Apply the Apriori algorithm to the dataset of transactions and identify all frequent k-itemsets.
(b) Find all strong association rules of the form: $X \wedge Y \rightarrow Z$ and note their confidence values.
5. For the transactions given below, consider confidence=60% and minimum support=30%. Identify large itemsets (L-Itemset) at L=3 with possible associations using A-priori algorithm and generate F-List using FP-Growth algorithm. [12] (76 Ash)

Transactions	Items description
T1	A, B, C, T, M, P, D, K
T2	A, B, T, P, D, K
T3	B, C, T, D, M, A, P
T4	A, C, T, M, D,
T5	A,C, D, K, M
T6	B, C, T

6. Identify the candidate, frequent item sets and association rules for the following transaction data using Apriori algorithm. Take minimum support = 20%, minimum confidence 80%. [8] (74 Ash)

TID	ITEMS
1	M1, M2, M5
2	M2, M4
3	M2, M3
4	M1, M2, M4
5	M1, M3
6	M2, M3
7	M1, M3
8	M1, M2, M3, M5
9	M1, M2, M3

7. Explain Apriori algorithm. Use Apriori to generate frequent item sets with support of 50% for the following transaction database. [10] (70 Ch)

TID	Items
1	ACD
2	BD
3	ABCE
4	BDF

8. How does Apriori Algorithm optimize the brute force approach for frequent item set generation? [10] (71 Ch)
9. What are association rules? How can spriori algorithm be used to generate association rules. [10] (71 Shr)

4.3 FP-Growth, FP-Tree

1. What is FP-Growth Algorithm? Explain FP-Growth Algorithm with example. [2+5] (80 Bh)
 |→ Explain FP-Growth algorithm with example. [8] (74 Ash)
 |→ Explain FP-growth algorithm in detail. [10] (71 Shr)
 |→ What are the advantages of FP growth method? Explain FP growth algorithm. [2+6] (75 Ash)
 |→ What is a Frequent item set? Explain FP growth method with example. [1+8] (72 Ka)
 |→ When do we use Association analysis? Explain FP-Tree with an example. [2+5] (81 Ba)
 |→ What do you mean by frequent Pattern growth, draw FP-tree with given tabular data. [4+4] (75 Ch)

TID	Items
01	f, a, c, d, g, i, m, p
02	a, b, c, f, l, m, o
03	b, f, h, j, o, w
04	b, c, k, s, p
05	A, f, c, e, l, p, m, n

2. What is limitation of Apriori algorithm compared to FP-growth? A database has 5 transactions, given in table below. Let min support = 60% and min confidence = 80%. [2+7] (81 Ba)

TID	Item_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

- i) Find all frequent itemsets using FP-growth.
 - ii) List all of the strong association rules (with support and confidence).
3. Consider the given transactional database from a grocery store. Use a support threshold of 33.34% and confidence threshold of 60% to compute the following: [4+4] (78 Bh)

Transcation ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

- a) Build a frequent pattern tree (FP-Tree). Show for each transaction how the tree evolves.
 - b) Use FP-Growth algorithm to discover the frequent itemsets from this FP-tree.
4. How does FP growth approach overcomes the disadvantages of Apriori algorithm. For the transaction data given in table generate FP-Tree. [2+8] (74 Ch)

Transaction ID	Item set
T1	Camera, Laptop, Pen drive
T2	Laptop, Pen drive
T3	Laptop, Mobile, Earphone
T4	Earphone, Mobile
T5	Camera, Earphone
T6	Laptop, Mobile, Earphone

5. What is FP Tree? How FP-growth algorithm eliminate the problem of Apriori algorithm? Construct the FP tree and find association rules for the following transaction database using FG- Growth algorithm. Support = 30% and confidence = 75%. [10] (73 Shr)

Transaction ID	Items
1	P,R,S
2	R,S,T
3	P,Q,R
4	P,R,S,T
5	P,S,T
6	P,Q,T
7	Q,S,T
8	Q,R,T

6. Write short notes on: FP-Tree.

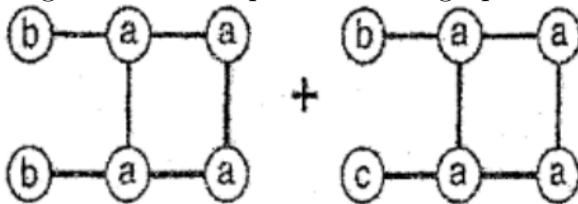
[5] (79 Bh)

4.4 Handling Categorical Attributes

- How do you handle the categorical attributes in data mining process? Explain with example. Generate the at least four subsequences from the given sequence: $\langle \{2,3,5\}, \{6,7,8\}, \{9,1\}, \{7,4\} \rangle$. [3+3] (79 Bh)
- What are Categorical data? What are the possible issues arrives when using Categorical data? How can you handle such issues? [2+3+3] (73 Shr)
- Write short notes on: Categorical data and related issues. [4] (72 Ch)

4.5 Sequential, Subgraph, and Infrequent Patterns

- (a) List all the 4-subsequences contained in the data sequence: $\langle \{1,3\} \{2\} \{2,3\} \{4\} \rangle$ [3+3] (76 Ch)
- (b) Draw all candidate sub-graphs obtained from joining the pair of graphs shown below using edge-growing method to expand the sub-graphs.



- What are subgraph pattern? [2] (79 Bh)
- Write short notes on: AprioriALL: Sequential pattern mining algorithm. [3] (76 Ash)
- Write short notes on: Sequential pattern. [3] (72 Ka)

5 Cluster Analysis

(9 Hours)

5.1 Basics and Algorithms

1. What is Cluster Analysis? What are its applications? Explain different types of clusters. [8] (81 Ba)
|→ What is a cluster analysis? How it is different from classification? [5] (72 Ka)
|→ What is the application of clustering in data mining? Explain K-means clustering with example. [2+6] (**75 Ch**)
|→ What is the application of clustering in data mining? Explain the k-means algorithm with example. [8] (73 Shr)
|→ When do we use clustering? How do you evaluate the cluster generated? [3+4] (**80 Bh**)
2. Why is a clustering an unsupervised learning? How can hierarchical clusters be generated using Bisecting K-means algorithm? [10] (**71 Ch**)
3. What are the types of clustering methods? Explain DBSCAN method of clustering with an example. [10] (**72 Ch**)
|→ List the various types of partition based clustering methods. Explain Hierarchical clustering method with an example. [10] (74 Ash)
4. What is contiguous cluster? Explain an algorithm that can be used to generate contiguous clusters. [10] (71 Shr)
5. An internet marketer is interesting in segmenting internet based the input attributes — top ten search key words used, top 10 URLs, recent 10 online purchases (vendor, product, qty, amt), Internet usage level, heaviest access hour, and heaviest access day of a week. Which clustering algorithm do you think can be used for segmentation? How do you validate the cluster which has been created? [2+6] (**79 Bh**)

5.2 K-means Clustering

1. Describe K-means algorithm for clustering and discuss strategy in determining the optimal value of K. [4+4] (80 Ba)
2. Use K-means clustering to cluster the following given data for $K = 2$ with Euclidean distance matrix. List down the demerits of this algorithm. [5+3] (81 Ba)
(The paper prints no data table for this question.)
3. Write K-means clustering algorithm. Generate two clusters from following dataset using K-means clustering. [2+6] (**78 Bh**)

Instance	A	B
1	1	2
2	2.5	1
3	3.5	1.5
4	4	1
5	3.5	2.5
6	5	3

4. Write K-means algorithm and find clusters for following data set. (Take $K = 2$) [2+8] (74 Ash)

Instance	X	Y
1	1.0	2.0
2	2.5	1.0
3	3.5	1.5
4	4.0	1.0
5	3.5	2.5
6	5.0	3.0

5. Explain K-means clustering with limitation. Generate two clusters from following dataset using K-means clustering. [4+6] (75 Ash)

A	B
1	2
2.5	4.5
4	6
3.5	4
4	5.5
3	6

6. Explain K-means clustering with limitation Use k-means clustering to cluster the following dataset. [10] (71 Shr)

A	B
1.0	1.0
1.5	2.0
3.0	4.0
5.0	7.0
3.5	5.0
4.5	5.0
3.5	4.5

7. Given the matrix (X) whose rows represent different data points, perform a k-means clustering on this dataset using the Euclidean distance as the distance function. Here (K) is chosen as 3. The center of the 3 clusters are initialized as red (6.2, 3.2), green (6.6, 3.7) and blue (6.5, 3.0). Provide the final cluster centers and comment on the number of iterations required for the clusters to converge. [8] (76 Ch)

$$X = \begin{bmatrix} 5.9 & 3.2 \\ 4.6 & 2.9 \\ 6.2 & 2.8 \\ 4.7 & 3.2 \\ 5.5 & 4.2 \\ 5.0 & 3.0 \\ 4.9 & 3.1 \\ 6.7 & 3.1 \\ 5.1 & 3.8 \\ 6.0 & 3.0 \end{bmatrix}$$

8. Describe the difference between Hierarchical and partitioning clustering. How K-means clustering is applied? Verify using example. [2+8] (74 Ch)

5.3 Hierarchical Clustering

1. What is hierarchical clustering? Use this clustering approach to draw dendrogram for given data points. [2+7] (80 Bh)

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

2. Cluster the following samples based on complete-linkage algorithm and draw the dendrogram. Using Euclidean distance. [8] (80 Ba)

Point	x Coordinate	y Coordinate
p1	0.40	0.53
p2	0.22	0.38
p3	0.35	0.32
p4	0.26	0.19
p5	0.08	0.41
p6	0.45	0.30

3. The table below is a distance matrix for six objects:

[4+4] (76 Ch)

	A	B	C	D	E	F
A	0					
B	0.12	0				
C	0.51	0.25	0			
D	0.84	0.16	0.14	0		
E	0.28	0.77	0.70	0.45	0	
F	0.34	0.61	0.93	0.20	0.67	0

- Show the final result of hierarchical clustering with single-link by drawing a dendrogram.
 - Show the final result of hierarchical clustering with complete-link by drawing a dendrogram.
4. What is Hierarchical Clustering? Differentiate between agglomerative and divisive approach of hierarchical clustering. Augment your answer with appropriate illustrative examples. [10] (70 Ch)

5.4 DBSCAN Clustering

- What are core, border and noise points? Write the algorithm of DBSCAN clustering and explain how it is useful in handling the noisy data. [3+5] (79 Bh)
 → How DBSCAN clustering is used for handling noise in data? [8] (75 Ch)
 → How DBSCAN algorithm works? How do we avoid the issues of DBSCAN? [8+2] (76 Ash)
 → Explain a DBSCAN algorithm with example. [7] (72 Ka)
- Provide answers to the following with regard to the DBSCAN clustering approach: [2+2+2] (78 Bh)
 - How does the DBSCAN quantify the neighborhood of an object? How is a large dense region assembled from small dense regions centered by core objects?
 - How does DBSCAN find clusters? How are the neighborhood threshold (Epsilon) and minimum number of points (MinPts) determined empirically in DBSCAN?
 - Prove that in DBSCAN, for a fixed minimum number of points (MinPts) value and two neighborhood thresholds, $\text{Epsilon}_1 < \text{Epsilon}_2$, a cluster (C) with respect to Epsilon_1 and MinPts must be a subset of a subset of a cluster (K) with respect to Epsilon_2 and MinPts.
- What is a density based cluster. Explain an algorithm that can be used to generate density based clusters. [8] (70 Ch)
 → What are outliers? Explain an algorithm that can be used to generate density based clusters. [8] (75 Ash)
- Write short notes on: Density reachable and Density Connected. [4] (74 Ash)

5.5 Issues: Evaluation, Scalability, Comparison

- Explain the different measures of cluster validity. [10] (71 Ch)
- How do you evaluate the cluster generated? (asked with “When do we use clustering?” — see 5.1) [4] (80 Bh)
- Write short notes on: Issues in clustering. [3] (74 Ch)
- Write short notes on: Cluster evaluation. [3] (72 Ka)

6 Anomaly / Fraud Detection

(3 Hours)

1. What is anomaly detection? Explain distance based method for anomaly detection. [2+3] (80 Ba)
|→What is anamoly detection? Explain distance based method for anamoly detection. [8] (73 Shr)
|→Why anamoly detection is important? Explain distance based method for anamoly detection. [2+6] (75 Ash)
|→What do you mean by anomaly detection and why is it important? Describe distance based approaches for anomaly detection. [4+3] (**74 Ch**)
|→What is an Anomaly Detection? Explain few distance based approaches that can be used for Anomaly Detection. [10] (**71 Ch**)
|→What is outlier? Explain the distance base approaches for the anomaly detection. [5] (**75 Ch**)
2. What is Anomaly Detection? Why is Anomaly Detection important? Briefly explain different types of anomaly detection schemes. [2+2+4] (81 Ba)
|→What do you mean by anomaly detection? Why is it important and where is it applicable? [3+3] (**79 Bh**)
|→What is an Anomaly detection? Discuss its importance in security. [5] (72 Ka)
3. Explain different types of outlier with suitable examples. How density based outlier detection works? [5+3] (**80 Bh**)
4. Compare and contrast among three difference methods of anomaly detection. [6] (**78 Bh**)
5. (a) Discuss the issues related to anomaly detection. [2] (**76 Ch**)
(b) If the probability that a normal object is classified as an anomaly is 0.01 and the probability that an anomalous object is classified as anomalous is 0.99, then what is the false alarm rate and detection rate if 99% of the objects are normal? [3] (**76 Ch**)
6. How can Nearest-Neighbor algorithm be used for anomaly defection? [10] (71 Shr)
7. Write short notes on: Clustering and its application in anomaly detection. [3] (76 Ash)
8. Write short notes on: Issues in anomaly/Fraud detection. [4] (**72 Ch**)
9. Write short notes on: Data Mining for Anomy Detection. [4] (74 Ash)
10. Write short notes on: Anomaly Detection; Base Rate Fallacy; Convex Hull Method. [15] (**70 Ch**)

7 Advanced Applications

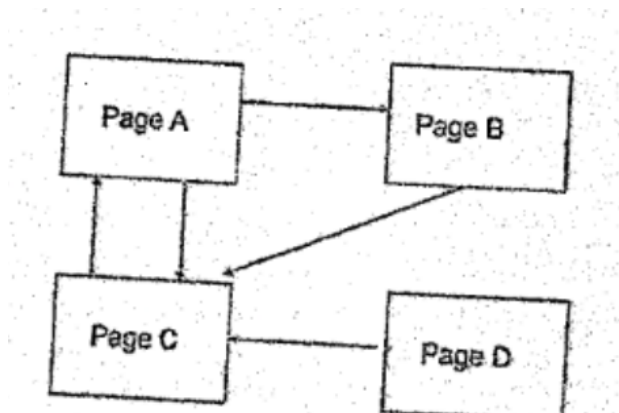
(3 Hours)

7.1 Mining Object and Multimedia

1. Explain Web mining and Multimedia mining. [6] (75 Ash)
2. Write short notes on: Multimedia mining. [3] (74 Ch)
3. Write short notes on: Visual Data Mining. [4] (75 Ch)

7.2 Web-mining

1. What is Web Mining? Briefly explain structure of Web Mining. [3+5] (80 Bh)
|→ What is web mining? Explain different categories of web mining. [6] (74 Ash)
|→ Explain web mining taxonomy. [8] (76 Ash)
|→ What are the challenges of web mining? [5] (75 Ch) — asked with time series, full statement in 7.3
2. Explain briefly the key steps in text mining. How do you find page rank? Explain. [4+4] (81 Ba)
3. Consider the following subset of pages and their links. Apply the PageRank algorithm using a damping factor of 0.85. A minimum of five iterations are required. Assume initial page rank of all pages is 0.25. [8] (76 Ch)



4. Write short notes on: Page Rank algorithm. [5] (79 Bh)
5. Write short notes on: Page rank algorithm in Web mining. [4] (78 Bh)
6. Write short notes on: Web mining. [4] (73 Shr) [3] (74 Ch) [-] (70 Ch)

7.3 Time-series data mining

1. Explain Time series data mining in brief. [6] (72 Ka)
2. What are the challenges of web mining? Explain about time series data mining with an example. [5] (75 Ch)
3. Write short notes on: Time Series Data Mining. [5] (80 Ba) [3] (74 Ch) [4] (72 Ch) [-] (71 Shr)